PROJECT TITLE: MULTI-VENDOR ONLINE MARKET PLACE

1. ABSTRACT

In many businesses, managing products, sellers, inventory, and customer orders can become Complex when multiple sellers operate on a single e-commerce platform. Traditional shopping systems may face challenges in maintaining accurate stock details, managing seller-specific orders, tracking order status, and providing secure access to users. This project, Multi-Vendor Online Marketplace with Order Workflow Engine, is developed using Java Spring Boot, MySQL, and HTML, CSS, and JavaScript. The system provides a centralized platform to manage product listings, customer purchases, seller information, and inventory efficiently.

The system provides three role-based portals: Admin, Seller, and Customer. The Admin manages user accounts, approves seller registrations, and monitors marketplace activities. Sellers can add products, update prices and stock quantities, and manage their orders. Customers can browse products, add items to their shopping cart, place orders, and track their purchases. The core highlight is the Order Workflow Engine, which automatically divides a customer's order into seller-specific suborders when products are purchased from multiple sellers. The system validates stock availability, calculates order totals, updates inventory, and maintains order status history. This makes the marketplace more organized, reliable, and efficient while reducing manual work and improving the overall shopping experience.

2. OBJECTIVES:

1. To develop a centralized online marketplace where multiple sellers can list and sell their products.

2. To provide secure registration and login for administrators, sellers, and customers.

3. To allow sellers to add, update, and manage products, prices, categories, and stock quantities.

4. To enable customers to browse products, search by name or category, and manage their shopping carts.

5. To implement a checkout system that validates product availability and calculates order totals automatically.

6. To divide orders into seller-specific suborders when a customer purchases products from multiple sellers.

7. To maintain accurate inventory and prevent orders from exceeding available stock.

8. To track order status and maintain a history of order updates.

9. To provide role-based access control so that users can access only the operations permitted for their roles.

10. To maintain order records and provide an organized platform for managing marketplace operations.

3. EXISTING SYSTEM:

In the existing system, sellers may use different websites or applications to sell their products. Managing products, stock, and orders separately can take more time. Customers may also need to visit different websites to buy products from different sellers.

Limitations:

1. Managing products from multiple sellers can be difficult when seller operations are not integrated.
2. Manual stock updates may cause inventory mismatches and overselling.
3. Customers may find it difficult to track orders purchased from different sellers.
4. Order processing and status updates may require unnecessary manual intervention.
5. Sellers may not have a dedicated interface to manage only their own products and orders.
6. Poor access control can expose customer or seller information to unauthorized users.
7. Maintaining accurate order history and inventory records can become complicated as the number of transactions increases.

4. PROPOSED SYSTEM

The proposed Multi-Vendor Online Marketplace with Order Workflow Engine provides a centralized web application where multiple sellers can list their products and customers can purchase products through a single platform.

How System Works:

1. User Registration and Login: Users register and log in to the system. The Admin can manage user accounts and approve seller registrations.

2. Product Management: Sellers can add new products, update product information, change prices, and manage stock quantities.

3. Product Browsing and Cart: Customers can browse and search for products, view product details, add items to the shopping cart, and update quantities.

4. Order Placement: When a customer places an order, the system validates product availability, calculates the total amount, and verifies the order details.

5. Order Workflow Engine: If the cart contains products from multiple sellers, the system creates a main customer order and separates it into seller-specific suborders.

6. Inventory and Order Tracking: The system updates stock after successful order placement. Sellers can manage order statuses, and customers can track order progress.

7. Admin Monitoring: The Admin can manage users, approve sellers, monitor products and orders, and oversee marketplace activities.

5. ROLES AND RESPONSIBILITIES

1. Admin:

- Manage customer and seller accounts.
- Approve or reject seller registrations.
- Monitor products listed on the platform.
- View and monitor customer orders.
- Manage the overall marketplace.

2. Seller:

- Register and log in to the seller account.
- Add new products and update existing products.
- Manage product prices and stock quantities.
- View orders received for their products.
- Update the status of their orders.

3. Customer:

- Register and log in to the website.
- Browse and search for products.
- Add products to the cart and place orders.
- View order details and track purchases.
- Check previous orders.

6. TECH STACK

Components	**Technology**
Back-end	Java, Spring Boot
Database	MySQL
Database connectivity	Spring Data JPA, Hibernate
Front-end	HTML, CSS, JavaScript
API Development	REST APIs
Security	Spring Security